

SHAMBHAVI PANDEY

B.E. Computer Science & Engineering • AI/ML & Full-Stack Software Engineer • Class of 2027
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SUMMARY

Final-year Computer Science student (CGPA 9.58/10.0) specializing in AI/ML, NLP, and full-stack software engineering, with hands-on experience building and shipping production-oriented machine learning systems, generative AI pipelines (RAG, LLM integration), and RESTful web applications. Proven track record leading cross-functional teams in hackathons, debugging complex ML pipelines to production-grade accuracy, and delivering end-to-end products from data pipeline through deployment.

EDUCATION

BNM Institute of Technology, Bangalore B.E. Computer Science and Engineering • CGPA: 9.58 / 10.0	<i>Aug 2023 – May 2027</i>
Wendy School Junior College, Gwalior 12th (ISC) — 86% • 10th (ICSE) — 96%	<i>Graduated 2023</i>

EXPERIENCE

Deep Learning & Reinforcement Learning Intern — E2E Management Services	<i>Jul 2025 – Aug 2025</i>
- Benchmarked CNN vs. transfer learning (VGG16) for image classification, improving test accuracy from 84.51% to 87.73% using ImageNet-pretrained feature extraction - Evaluated RNN, LSTM, and GRU architectures on sequence-modeling tasks; tuned a simple RNN to 99.65% validation accuracy, outperforming LSTM (86.64%) and GRU (80.58%) - Optimized CNN generalization via dropout, L2 regularization, learning-rate scheduling, and activation tuning, raising test accuracy from ~97.2% to 98.3% - Conducted a fairness/bias audit across demographic subgroups on a production classifier, surfacing accuracy gaps of up to 6 percentage points to inform model remediation	
Full-Stack Software Engineering Intern — BNM Institute of Technology	<i>Aug 2024 – Oct 2024</i>
- Shipped 3 responsive, production-ready web applications using React, Node.js, and MongoDB; improved page-load performance via lazy loading and optimized API payloads - Designed and implemented RESTful APIs for authentication and data persistence; validated 15+ endpoints end-to-end with Postman across dev and staging environments - Collaborated in a 2-person Agile team through sprint planning, code review, and cross-functional debugging, shipping all features on schedule	

PROJECTS

- Diagnosed and fixed a critical data-leakage bug in the training pipeline that had collapsed model accuracy to 0%, retraining a clause classifier on 12K+ labeled spans (CUAD dataset, 41 categories) to reach 71% test accuracy
- Root-caused and resolved 3 additional defects through systematic debugging — a label-vocabulary mismatch blocking clause detection, a missing similarity-check feature, and a logic error discarding extracted text

Diabetes Risk & Guidance System — Hybrid AI Clinical Decision Support | TensorFlow, Flask, RAG (FAISS), Azure OpenAI

- Engineered a hybrid AI pipeline where a TensorFlow neural network (72.7% test accuracy, PIMA dataset) predicts diabetes risk, then a FAISS-based RAG layer grounds an Azure OpenAI (GPT-4o-mini) call to generate a structured, evidence-based patient report
- Built a Flask backend orchestrating the full workflow — prediction, retrieval, LLM generation, PDF creation (ReportLab), and agent-based email delivery (Semantic Kernel) — plus a chat endpoint that re-embeds each report into FAISS for interactive follow-up Q&A

EchoGuard — Real-Time AI Speech Analysis for Mental Health | Python, NLP, Audio ML, Signal Processing

- IBM MindScope Hackathon Finalist (top teams, 200+ competing): built a real-time speech-analysis pipeline detecting indicators of suicidal ideation from audio using prosodic feature extraction and NLP classification
- Designed the full ML architecture — data ingestion, feature pipeline, model training, and evaluation — with custom audio preprocessing for noise reduction, achieving competitive classification metrics on held-out test sets

Additional Projects: MAITRI — multimodal (CV + audio + NLP) stress/emotion assessment AI, Smart India Hackathon 2025 • Explore India — React travel platform with login-gated UI • Blockchain TextFile Storage — Java, hash-chain integrity • SlumberVista: DreamLens — NLP-based dream journal analysis with mood classification

LEADERSHIP & POSITIONS OF RESPONSIBILITY

- Hackathon Organizer & Co-Lead, TATVA 2026** — BNM Institute of Technology 2026
- Spearheaded a college-wide hackathon end-to-end for 200+ participants — registrations, scheduling, judging logistics — coordinating a volunteer team to deliver a zero-downtime event
- Head Girl** — Wendy School Junior College, Gwalior 2021 – 2023
- Elected highest student representative; led and mentored a cohort of 800+ students, bridging student–administration communication and driving disciplinary governance

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, SQL
- AI / ML:** TensorFlow, scikit-learn, NLP, Computer Vision, Reinforcement Learning, Generative AI, Retrieval-Augmented Generation (RAG), Gemini API, Azure OpenAI
- Web & Backend:** React, Node.js, FastAPI, Flask, REST APIs, Firebase, MongoDB, MySQL
- Tools & Platforms:** Git, GitHub, Postman, VS Code, Linux, Docker (familiar), HuggingFace (familiar)
- Concepts:** Data Structures & Algorithms, OOP, System Design (fundamentals), Agile/Scrum
- Problem Solving:** 100+ LeetCode problems solved across arrays, dynamic programming, graphs, and string manipulation

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL — Introduction to Internet of Things: 90/100 (Elite)
- IBM SkillsBuild • Intel Diya Program for Data Analytics • MongoDB Developer Course (MongoDB University)
- IBM MindScope Hackathon — Finalist (200+ participants) • Google GenAI Exchange Hackathon — Project Lead
- SAP Hackfest 2025 (NextGrids) • NEC EUREKA (IIT Bombay E-Cell) • NSS Active Member (2024–Present)